Course Outline & Introduction

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Introduction to Statistics (STAT 1200)

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THOMPSON RIVERS UNIVERSITY

Topics

2. Land Acknowledgment
3. Introduction
4. Lectures
5. Grades
6. Assignments
7. Tests
8. Project
9. How to get Help with Coursework

Thompson Rivers University campuses are on the ancestral lands of the Tk'emlúps te Secwépemc (Kamloops campus) and the T'exelc (Williams Lake campus) within Secwepemcúlecw, the traditional and unceded territory of the Secwépemc. The region TRU serves also extends into the territories of the St'át'imc, Nlaka'pamux, Nuxalk, Tšilhqot'in, Dakelh, and Syilx peoples.

Course Basics

- Course: Introduction to Statistics (STAT 1200)
- Lecturer: Ing. Sean Hellingman, Ph.D.
- Meeting times:
  - Lectures: Tues/Weds 17:00 - 18:15 OM 1340
  - Seminars: Mon 16:00 - 17:15 in OM 1330

Course Educational Outcomes I

1. Summarize and describe data using appropriate numerical, tabular, and graphical methods, as well as comprehend and appraise the use of such summaries.

2. Identify data types and corresponding related theoretical distributions

3. Demonstrate understanding of basic probability concepts

4. Design, communicate, and appraise appropriate statistical experiments based on an understanding of the study question and type of data. Understand the limitations of experiments.

Course Educational Outcomes II

5. Employ, interpret the results of, communicate, and appraise appropriate inferential methods such as statistical tests and estimation procedures based on an understanding of the study question, type of study design and type of data. Determine potential fallacies, errors, and limitations in conclusions.

6. Comprehend, communicate, and appraise the use of statistical terms and practices in appropriate concepts to convey concepts associated with probability, sampling, and inference.

7. Students learn how various methods applied to the same data can provide similar or contrary results even when used appropriately.

Tentative Topics

- Displaying and Summarizing Data
- Probability
- Design
- Sampling and Estimation
- Hypothesis Testing
- Regression Analysis
- *R Studio*

Software

- R within the R Studio environment.
  - *Excel and Python will be accepted for some topics.*
- The purpose of this is to simplify calculations and encourage reproducible science.

Moodle

- All important course information will be located on the course moodle page.
  - Course information and announcements
  - Lecture slides
  - Examples
  - Assignments
  - Seminar Resources
  - Grades

Lectures

- Content is focused on the applied side of statistical theory.
- Combination of lecturing and working through examples.
- Please ask questions as they arise!
- Practice *exercises* are included at the end of the lecture slides for your convenience.
  - Further examples will be presented in the seminars.
- **Expectations:**
  - Please be respectful to everyone.
  - **Please don't be disruptive to your colleagues.**
  - Please leave your mobile devices on silent.

Lecture Participation

- We are in a classroom setting and I encourage you to bring your laptops to work through the examples.
- I will post example code **prior** to lectures and seminars.
- It is strongly encouraged to work along through the examples.

Seminars

- We will work through additional examples related to the content covered in the lectures.
- We are in a computer lab & R is loaded on the computers.
- Any remaining time will be dedicated to homework help.
- You are encouraged to ask questions and work through the examples.

Course Evaluation Weights

- Assignments 30%
- Tests 40%
- Project 30%
  - 20% Written report
  - 10% Oral presentation

TRU Letter Grades

Letter Grade	Numerical Grade	Grade Points	Letter Grade Definitions
A+ **A** **A-**	90–100 85–89 80–84	4.33 4.00 3.67	Excellent. Superior performance showing comprehensive, in-depth understanding of subject matter. Demonstrates initiative and fluency of expression.
B+ **B** **B-**	77–79 73–76 70–72	3.33 3.00 2.67	Very good. Clearly above average performance with knowledge of principles and facts generally complete and with no serious deficiencies.
C+ **C**	65–69 60–64	2.33 2.00	Satisfactory. Basic understanding with knowledge of principles and facts at least adequate to communicate intelligently in the discipline.
C-	55–59	1.67	Pass. Some understanding of principles and facts but with definite deficiencies.
D	50–54	1.00	Minimal pass. A passing grade indicating marginal performance. Student not likely to succeed in subsequent courses in the subject.
F	0–49	0.00	Unsatisfactory. Fail. Knowledge of principles and facts is fragmentary.

Assignments (30%)

- 5 total assignments:
  - Focus on practical applications based on the lectures.
  - Assignments will be posted on moodle with due dates clearly specified.
  - Due dates can be found in the tentative calendar & on moodle.
  - The **lowest** grade will be dropped.
  - Collaboration is **encouraged** but please **submit separate and *different* documents.**
  - **Note:** *Due dates are not always consistent.*

Assignment Expectations (30%)

- **Expectations:**
  - Assignments are mandatory in this course.
  - Assignments will be submitted (unless otherwise instructed) as PDF files in moodle.
  - Please include your name and TRU ID on your assignment.
  - ***Late assignments will not be graded.***
  - *LLMs can help when you are stuck, they are not to be used to complete assignments.*
    - *The student may be penalized for any solutions that appear to be directly produced by generative AI.*

Tests (40%)

- 2 total tests:
  - Test I is tentatively scheduled for **02 March during the seminar period**.
  - Test II will take place during the exam period.
  - More details about the tests will be uploaded closer to the test dates.

- **Expectations:**
  - Please leave personal items, including mobile phones, in your bag at the front of the class.
  - Please bring and be ready to present English/French language identification (TRU student card) to the tests.
  - **The TRU Academic Integrity policy is always active.**

Project (30%)

- 1 term project:
  - Designed to apply statistical methods to real-life data.
  - Groups of 1-4 students may work on the same project.
  - Tentatively due the last week of classes.
  - Presentations will be scheduled for the last two weeks of classes.
  - *More detailed expectations will be posed soon.*

Office Hours

- Room: CT412
- Times:
  - Tuesday 16:00 - 16:50
  - Wednesday 15:45 - 16:35
  - *Or by appointment*

Email

- Email questions to: shellingman@tru.ca
- I will be happy to help when I can!
- **Email Expectations:**
  - Please, include an informative subject.
  - Please, clearly state your exact problem and what you have already tried.
  - Please do not email questions last minute (you may not get help in time).

Suggested Textbooks

- The lectures, exercises, seminars, and assignments should be sufficient.
- I will also post links to open source resources throughout the course.
- If you want more information on these topics:
  - Kohl, K., (2022). *Introduction to statistical data analysis with R (Second Edition)* Retrieved from https://github.com/stamats/ISDR/blob/main/IntroductionToStatisticalDataAnalysisWithR_ed2.pdf.
  - Devore, J. L., Berk, K. N., & Carlton, M. A. (2012). *Modern mathematical statistics with applications (Second Edition).* New York: Springer.
  - Evans, J. R., Olson, D. L., & Olson, D. L. (2007). *Statistics, data analysis, and decision modeling.* Upper Saddle River, NJ: Pearson/Prentice Hall.

Student Resources at TRU

- Academic Supports
- Health and Wellness
- Diversity & Equity
- Career & Experiential Learning
- Security
- Students' Union

TRU SAFE App

- App: https://www.tru.ca/risk-management-services/security/tru-safe-app.html
- Apple
- Android

Accessibility

- All TRU students who require accommodations are encouraged to register with Accessibility Services upon registering with TRU.
- Help determine how test and exam accommodations can be arranged.

AI

LLM AI (ChatGPT)

- "*It works best for tasks we could do ourselves but shouldn't waste time on, yet it can actively harm our learning when we use it to skip necessary struggles*" - Ethan Mollick
- **The instructor reserves the right to question and penalize the student on any solutions that appear to be directly produced by generative AI.**

Course Outline

- Please refer to the official Course Outline document for additional information about the course.

Exercise 1

- Select a topic that you are knowledgeable about:
  - Ask ChatGPT to write a 3 paragraph essay about this topic.
  - How accurate is the essay?
  - Does it sound natural?
  - Would you be comfortable using this essay to teach someone who knew nothing about this topic?